RA-ClipScore: Making Generative Model Evaluation More Interpretable

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Motivation

Generative models produce increasingly realistic images, yet existing evaluation metrics offer limited interpretability.

- **Traditional metrics (FID, KID):** Reduce performance to a single score, without explaining why one model outperforms another.
- **CLIP-based metrics:** Measure global image-text alignment but provide limited insights into which attributes or spatial regions contribute to performance differences.

Our Goal: Move beyond single scores to interpretable evaluation that reveals why and where models differ.



Methodology

RA-CLIPScore enables fine-grained, attribute-wise and spatial evaluation through three key components:

1. **Dual Prompts:** Evaluate each attribute independently using positive and negative prompts, reducing competition among attributes.
2. **Regional Feature Extraction:** Preserve local patch representations while retaining global semantics, enabling fine-grained spatial understanding.
3. **Attention Refinement & Region-Aware Aggregation:** Refine patch-level scores using attention and aggregate them into robust attribute-wise scores.

From Attribute Scores to Distribution Evaluation:

- **SaD:** Detects individual attribute distribution shifts.
- **PaD:** Captures changes in attribute co-occurrence patterns.
- **R-SaD:** Reveals spatial biases across image regions.

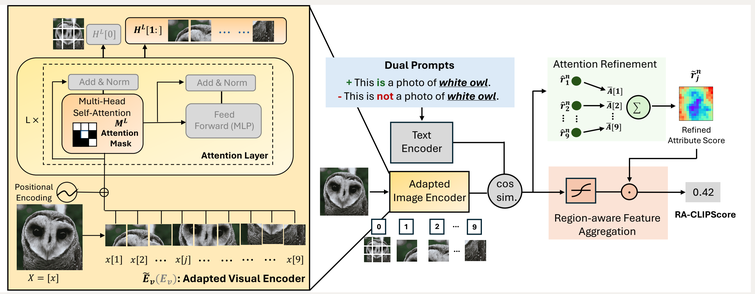
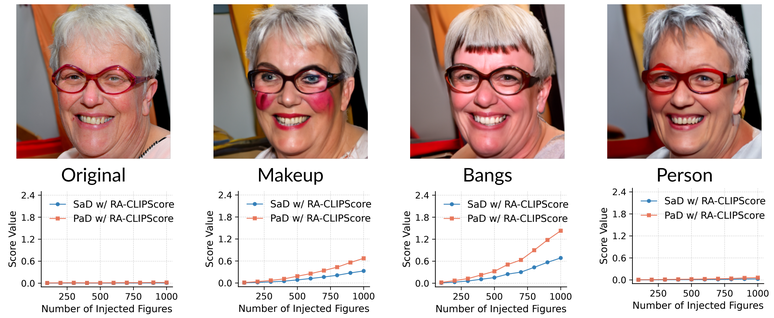


Figure 1. Our proposed RA-CLIPScore. We introduce dual prompts and regional feature extraction to improve interpretability of CLIP-based evaluation metrics while extending to spatial domain. Dual prompts decouple correlations among attributes, whereas regional feature extraction enhances fine-grained sensitivity through an adapted visual encoder, attention refinement, and region-aware feature aggregation. Grey blocks (left) denote components unchanged from the original visual encoder.

Injection Experiment

- **Experiment:** Progressively inject attribute-biased images (makeup and bangs) into FFHQ.
- **Finding:** SaD and PaD increase with stronger attribute biases, while remaining stable for control conditions.



Classification Experiment

- **Experiment:** Evaluate attribute classification robustness by introducing irrelevant attributes alongside relevant image–attribute pairs.
- **Finding:** RA-CLIPScore achieves the best accuracy and F1 scores.

Method	Acc. ↑	Macro-F1 ↑	Micro-F1 ↑
HCS	79.30	42.12	54.20
CLIPScore	79.44	44.04	54.52
RA-CLIPScore	**79.75**	**45.03**	**55.20**

Dataset Comparison Experiment

- **Experiment:** Compare datasets with reversed gender-eyeglasses dependencies. Dataset C reverses the attribute coupling in A and B.
- **Finding:** RA-CLIPScore shows stronger attribute sensitivity and disentanglement.

Method	Datasets	SaD_{EG}	$\text{SaD}_{\neg EG}$	PaD
RA-CLIPScore	A vs. B	4.52	1.11	2.80
	A vs. C	693.97	1.10	79.00
HCS	A vs. B	3.89	3.43	20.97
	A vs. C	93.86	5.94	54.98

Ablation Study

- **Experiment:** Evaluate the contribution of each component on 20 attributes by measuring positive–negative score separation using Cohen's d (↑).
- **Finding:** The full RA-CLIPScore achieves the best average discriminability across all attribute groups. Global embeddings benefit global attributes, while local tokens are essential for local attributes. Dual prompts and attention refinement further improve attribute separation.

Attribute	CLIPScore	HCS	Ours (Full)	w/o DP	w/o LT	w/o GE	w/o AR
Global (11)	0.83	0.72	**1.13**	0.76	1.11	0.90	0.83
Local (5)	1.47	1.75	**2.03**	1.73	1.62	1.87	1.89
Other (4)	0.80	0.73	**0.92**	0.76	0.89	0.75	0.72

Table 1. Ablation study on 20 CelebA attributes grouped into three categories. Higher mean Cohen's d indicates better attribute separation (↑). DP: dual prompts; LT: local tokens; GE: global embeddings; AR: attention refinement.

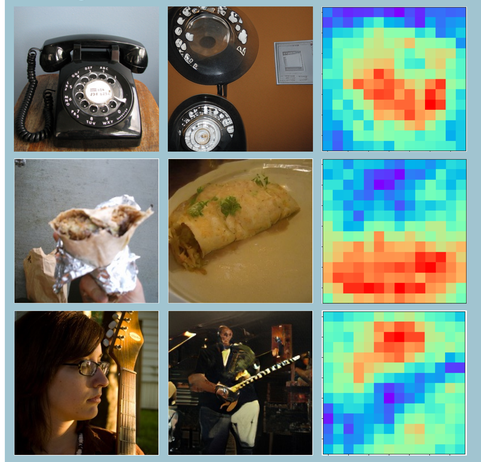
Evaluation of Generative Models

- **Experiment:** Compare 50K generated images per model against 50K real FFHQ images using 20 facial attributes.
- **Finding:** SaD and PaD show rankings consistent with traditional quality metrics, while revealing attribute-level discrepancies. StyleGAN2 performs best, whereas LDM exhibits larger biases, particularly in *bald head*.

Model	FID ↓	KID ↓ ($\times 10^{-4}$)	Prec. ↑	Recall ↑	Coverage ↑	Density ↑	SaD ↓ ($\times 10^{-3}$)	PaD ↓ ($\times 10^{-3}$)
StyleGAN2	**2.94**	**5.0**	**0.792**	0.665	**0.948**	**1.11**	**2.25**	**4.9**
StyleGAN3	5.20	8.2	0.769	**0.700**	0.947	1.02	2.98	6.8
LDM (50)	8.85	60	0.703	0.666	0.832	0.72	5.64	12.3
LDM (200)	9.53	69	0.700	0.663	0.827	0.67	6.44	14.0

Spatial Evaluation of Generative Models

- **Experiment:** Compare generated and real ImageNet images across 1,000 classes using R-SaD to identify spatial distribution discrepancies.
- **Finding:** R-SaD reveals distinct model-specific spatial biases:
  - **StyleGAN-XL:** Rotary dial phones appear at more varied locations.
  - **BigGAN:** Electric guitars favor a diagonal orientation ($\sim 45°$).
  - **LDM:** Burritos appear more frequently in upper regions.



Quantitative Results

Model	R-SaD ↓
StyleGAN-XL	$\mathbf{0.0136 \pm 0.0140}$
BigGAN	0.0232 ± 0.0234
LDM (250)	0.0146 ± 0.0147

Top-3 Discrepant Classes

Model	Classes
StyleGAN-XL	Rotary dial telephone Carousel iPod
BigGAN	Balloon Electric guitar Microphone
LDM (250)	Burrito Chain-link fence Carton

Figure 2. Spatial evaluation on ImageNet. Left: real images, generated samples, and R-SaD heatmaps (StyleGAN-XL, LDM, BigGAN from top to bottom). Red indicates higher occurrence in real images, while blue indicates higher occurrence in generated images.

References

[1] Y. Lu, T. Kucherenko, H. Kjellström, and J. Bütepage, "Ra-clipscore: Making generative model evaluation more interpretable," in *Neural Information Processing Systems*, 2026. [Online]. Available: https://arxiv.org/abs/2608.12088